



Results-driven Full Stack & AI/ML Engineer with extensive experience in machine learning and software development. Proficient in building scalable applications and optimizing models for high-dimensional data. Demonstrated ability to lead cross-functional teams and deliver innovative solutions, ensuring alignment with industry trends and client needs.

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SKILLS

- MACHINE LEARNING & AI : TensorFlow, PyTorch, Scikit-learn, Keras, JAX, Python, NLP/GPT, Computer Vision, Model Deployment, MLOps, Anomaly Detection, Feature Engineering, Signal Processing
- FULL-STACK DEVELOPMENT : React, Node.js, TypeScript, Python, JavaScript, C#/ASP.NET, Express.js, REST APIs, Database Design, Microservices
- CLOUD & INFRASTRUCTURE : AWS, Azure, GCP, Docker, Kubernetes, CI/CD, Terraform, PostgreSQL, DynamoDB, S3
- DATA & SCIENTIFIC COMPUTING : Spark/PySpark, SQL, Pandas, BigQuery, MATLAB, C++, OpenMP, MPI, Distributed Computing
- SPECIALIZED DOMAINS : Medical AI, Computer Vision, HPC, Hyperspectral Imaging, Finite Element Analysis

LANGUAGES

- English | Native

Andrew Jackson



EXPERIENCE

- Apr 2025
Jul 2026

MACHINE LEARNING ENGINEER
HV Vision. Carlsbad City, CA

Developed ML pipeline for hyperspectral imaging predictions. Engineered models for high-dimensional feature spaces while maintaining physics-based validity. Collaborated across diverse stakeholder groups including farmers, agronomists, hardware engineers, and executive leadership.
- Feb 2024
Sep 2024

TECH LEADER & SENIOR SOFTWARE ENGINEER
Yuga Labs. Coral Gables City, FL

Architected E2E Twitter bot for fraud detection on company brand impersonation; integrated Google ecosystem for legal team operations. Rearchitected backend infrastructure from key-value storage (Cloudflare Workers) to DynamoDB + S3 automated API. Built Web3 smart contract integrations enabling customer interactions with blockchain protocols.
- Mar 2022
Nov 2022

MACHINE LEARNING & AI EXPERT
Celegence LLC. Downers Grove City, IL

Identified client problems and proposed AI solutions with MVP demonstration. Architected end-to-end AI pipeline from data acquisition through production prediction. Set up MLOps infrastructure for client teams; developed question-answering systems for systematic document review. Built automated document inclusion/exclusion pipeline for document screening. Led agile weekly team meetings and positioned AI product for life science industry market entry.
- Jan 2020
Jun 2021

MACHINE LEARNING & DATA SCIENCE
Orlando Health. Orlando City, FL

Developed data and ML pipeline for ingesting massive healthcare datasets (financial and discharge records). Built predictive models for insurance, readmission, and clustering/segmentation analysis. Collaborated with multidisciplinary teams; prepared reports and presentations. Leveraged GCP compute engine for big data and model development; used PySpark for statistical analysis.
- Jul 2018
Mar 2025

SOFTWARE ENGINEER INSTRUCTOR
Codecademy. New York City, NY

Led in-depth courses on full-stack topics including Python, SQL, NoSQL, React, Redux, HTML, CSS, Go. Graded coursework and taught students scalable, well-architected code design. Mentored hundreds of students across diverse programming topics.
- Mar 2016
Mar 2018

SENIOR RESEARCH SCIENTIST, HPC SOFTWARE DEVELOPER
Hexagon Manufacturing Intelligence. North Kingstown City, RI

Developed ML models replacing computationally heavy simulations for auto parts design and optimization. Engineered high-performance parallel linear system solver achieving 1000x speedup using sparse direct solvers and iterative methods (PCG, algebraic multigrid). Built sparse/dense C++ linear algebra library with scientific code optimization. Developed probabilistic uncertainty propagation analysis tool. Authored and published peer-reviewed paper on spring back phenomenon.
- Sep 2015
Feb 2016

SOFTWARE ENGINEER INTERN
Starbucks. Seattle City, WA

Built live KPI dashboard scraping metrics from Starbucks database and displaying daily changes to C-level executives. Created live sentiment dashboard streaming tweets, applying NLP for sentiment analysis, and computing daily averages to track public perception over time.
- Feb 2014
Jul 2015

RESEARCH SCIENTIST
Alstom Power. Wexford, Allegheny County, PA

Developed semi-analytical software tool for improving old hydraulic turbine performance and avoiding costly replacements. Built predictive tool identifying dangerous structural responses to loads and fluid dynamic forces. Designed and validated experimental setups. Authored documentation, design guidelines, and internal white papers.
- Mar 2008
Feb 2013

PHD RESEARCHER
University California San Diego. San Diego City, CA

Developed mathematical models predicting complex nonlinear systems behavior (fluid-structure interaction, cardiovascular dynamics, weather forecasting). Built high-performance numerical software for solving nonlinear differential equations. Managed supercomputing cluster pipelines; processed and analyzed terabyte-scale results using OpenMP and MPI. Designed and performed experiments validating software predictions. Developed ML meta-models for forecasting without heavy simulations. Taught numerical methods, programming, mathematical modeling courses. Published results in journals and conferences.

EDUCATION

- Feb 2004
Dec 2008

BACHELOR OF COMPUTER SCIENCE
UC San Diego . San Diego City, CA

CERTIFICATES & COURSES

Kubernetes Application Developer (CKAD)
Deep Learning Specialization
TensorFlow Developer
Google Cloud Professional ML Engineer
AWS Certified Solutions Architect
AWS Certified Machine Learning
Azure Data Scientist Associate
Natural Language Processing
Professional Engineer (PEng)

